

Identifying Post-COVID Risk Factors with Model-Agnostic Feature Importance

using the `xplainfi` R package on the German national cohort (NAKO)

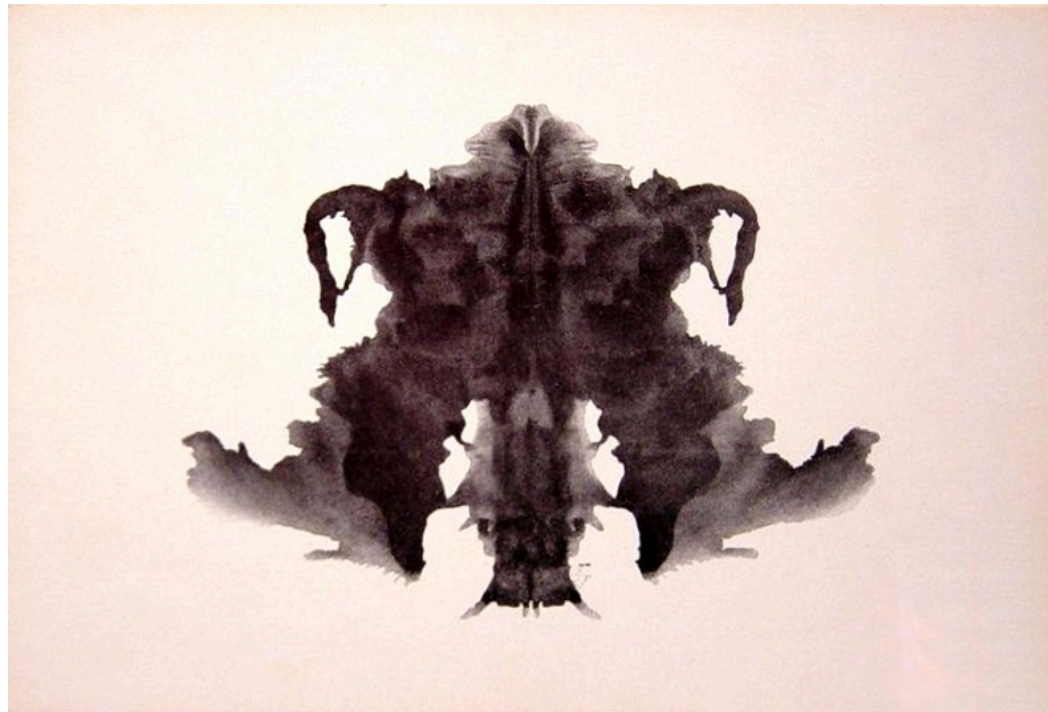
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2026-05-19

CEN 2026

A matter of interpretation



Post-COVID condition (PCC)

“Long- / Post-COVID”



2

- Persistent / recurring symptoms long after SARS-CoV-2 infection
- Clinically heterogenous (symptoms & severity)

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Federal Ministry of
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- “Importance” itself is not a single quantity $\rightarrow$ *method choice* defines the answer

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Permutation feature importance (**PFI**)

Conditional FI

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 - Conditional Gaussian $\rightarrow$ fast, but only continuous data
 - Adversarial random forests (ARF)
 - $\rightarrow$ handles *mixed* data, *missing values*, computationally more expensive

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Fixed model, learning algorithm, data-generating process?

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 - **LOCO** automatically refits, includes learner variability
 - **Any method:** Repeat across resampling (e.g. 15 subsampling iters)
 - Captures variability from data sampling and learner stochasticity

Analysis dataset: NAKO

German National Cohort



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- $N \approx 66250$ participants reporting ≥ 1 SARS-CoV-2 infection
- $\approx 24\%$ classified as **“any PCC”** (“Severe PCC” much rarer $\rightarrow$ not shown today)

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- Mix of *survey* + *in-person assessment*:
 - demographics, anthropometrics, socioeconomic
 - comorbidities, smoking history, biomarker labs (where available)
 - mental-health questionnaires
- ≈ 150 features used (dropping near-constant / near-fully-missing)
- Rare-event indicators retained despite low prevalence

Analysis: 'Any PCC' prediction

Preliminary, exploratory



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- Learners: Gradient boosting (XGBoost), random forest (ranger)
- Tuned on **PR-AUC** (baseline = 24%)
- General performance:
 - PR-AUC $\approx$ 43%
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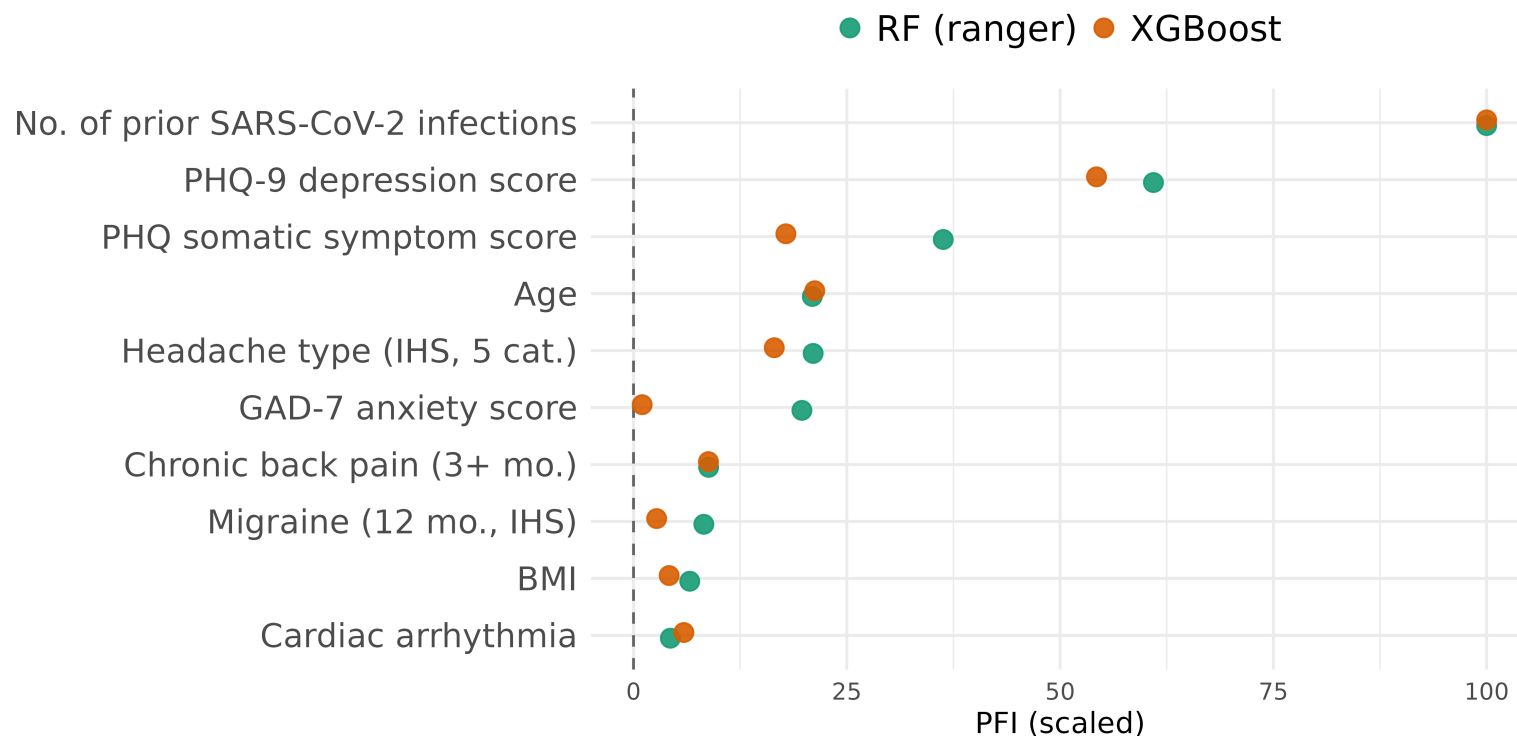


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- FI here on test set (i.e. **model importance**)

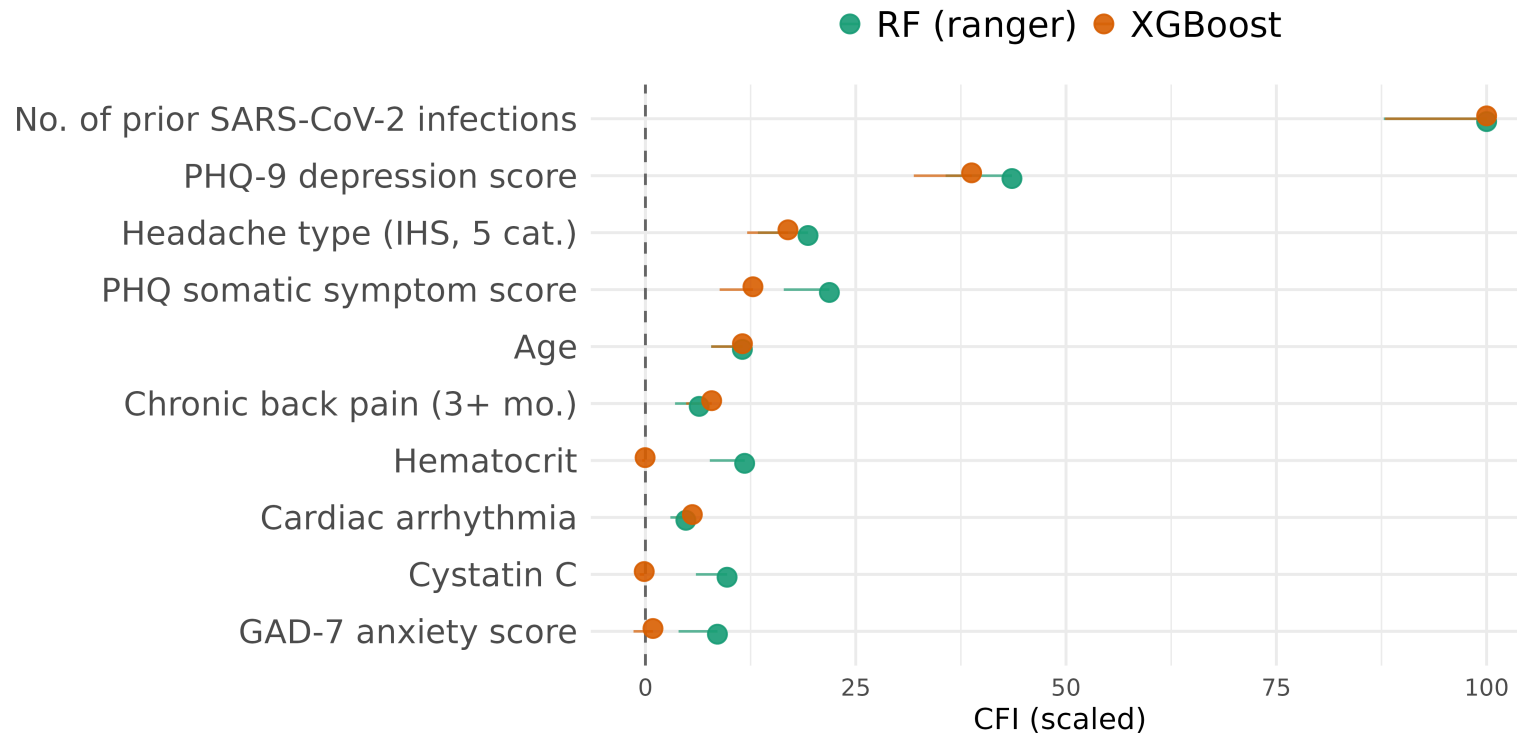
Results: Permutation Feature Importance (PFI)

Top 10 features (between RF + XGBoost)



Results: Conditional Feature Importance (CFI)

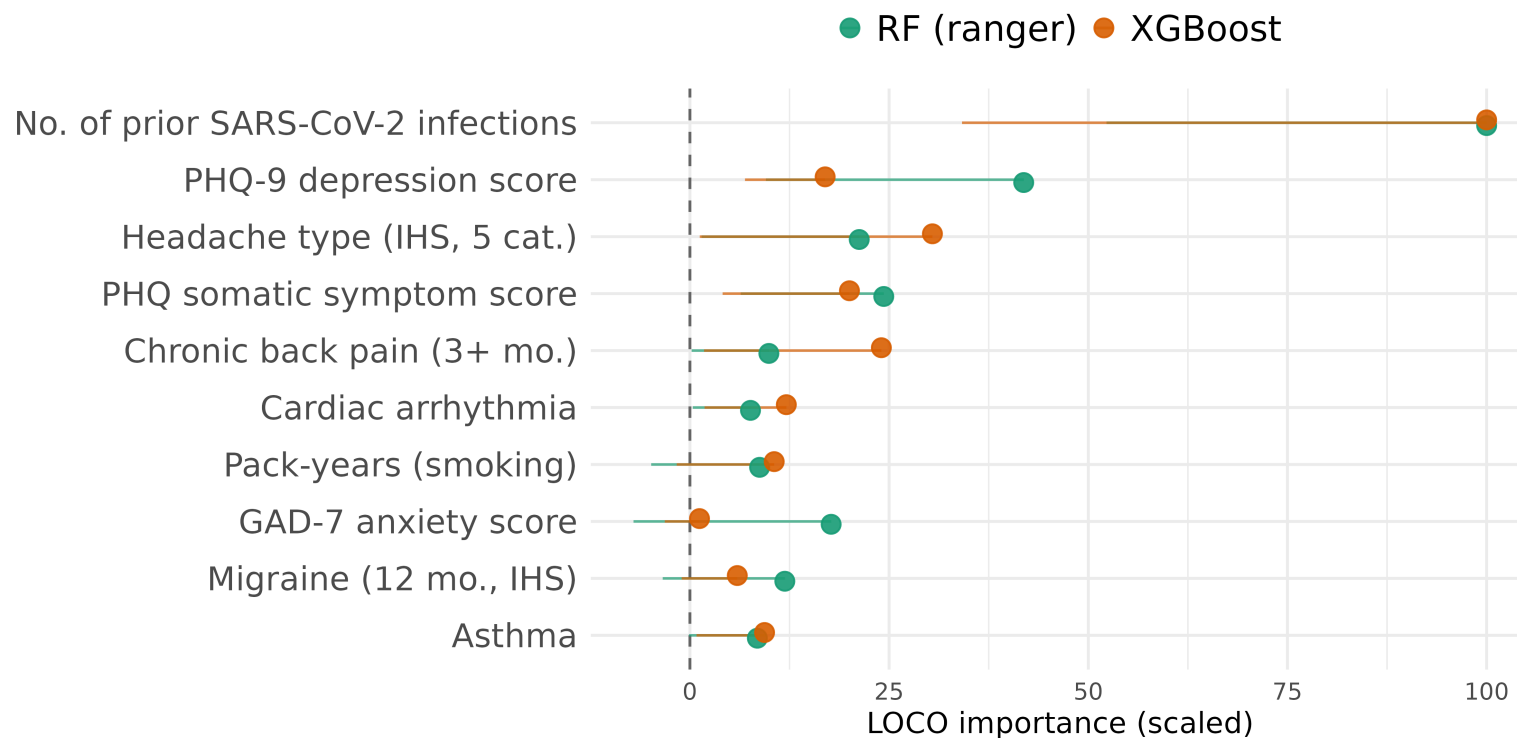
Top 10 features, one-sided 95% CIs



Results: Leave-one-covariate-out (LOCO)

Top 10 features, one-sided 95% CIs

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Conclusion

Everything is complicated, always



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- Feature importance is useful, but **not a single number**
- Each method answers a slightly different question
 - **PFI**: marginal, model-faithful but extrapolation issue
 - **CFI**: conditional, more faithful to joint distr., sampling-dependent
 - **LOCO**: refit-based, expensive but assumption-light

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 - **PFI**: marginal, model-faithful but extrapolation issue
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 - **LOCO**: refit-based, expensive but assumption-light
- Compare methods → robustness check, not contradiction
- `xplainfi` provides a unified interface for all of the above (Burk et al., 2026)

Thank you for your attention!



Contact

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References

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